

# Skills Summary

## Complex Solution Sets

Use this reference while completing your worksheet or when studying.

---

### Skill 1 — Square Roots of Negative Numbers

---

**Rule:**  $\sqrt{-k} = i\sqrt{k}$  for  $k > 0$ , and a square has *two* roots:  $x^2 = -k$  gives  $x = \pm i\sqrt{k}$ .  
Isolate the square first; divide out any leading coefficient before taking roots.

**Example:** Solve  $x^2 + 36 = 0$

**Step 1.** The square is already alone on one side after one move, so there is no need for the quadratic formula — isolate and take both roots.

**Step 2.**  $x^2 = -36$ , and the two square roots of  $-36$  are  $6i$  and its negative.  
 $\Rightarrow x = 6i$  or  $x = -6i$

**Try it:** Solve  $2x^2 + 98 = 0$

check:  $-x$  or  $x = x$

### Skill 2 — When the Discriminant Is Negative

---

**Formula:**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When  $b^2 - 4ac < 0$  the two solutions are non-real conjugates. Divide *both* parts of the numerator by  $2a$ .

**Example:** Solve  $x^2 - 6x + 25 = 0$

**Step 1.** There is no factor pair for 25 that sums to  $-6$ , so go to the quadratic formula and read the discriminant first.

**Step 2.** The discriminant is  $36 - 100 = -64$ , so the numerator is  $6 \pm 8i$ ; dividing both parts by 2 gives real part 3 and imaginary part 4.  
 $\Rightarrow x = 3 \pm 4i$

**Try it:** Solve  $x^2 + 2x + 26 = 0$

check:  $1 - x = x$

### Skill 3 — All Solutions of a Higher Degree

**Rule:** an equation of degree  $n$  has exactly  $n$  solutions, counted with multiplicity. Count what you have before you stop.

Factor into quadratics, then solve each one — a factor like  $x^2 + 4$  contributes a non-real pair, and it is the factor most often abandoned.

**Example:** Solve  $x^4 - 16 = 0$

**Step 1.** Degree four means four solutions, so factoring as far as possible is the way to find all of them.

**Step 2.**  $(x^2 - 4)(x^2 + 4) = 0$ : the first factor gives the real pair, and the second gives  $x^2 = -4$ , whose roots are imaginary.

$$\Rightarrow x = 2, -2, 2i, -2i$$

**Try it:** Solve  $x^4 - 5x^2 - 36 = 0$

check:  $x = 2, -2, 3i, -3i$

### Skill 4 — Checking and Reporting the Full Set

**To check a solution:** substitute it in and simplify. If the expression comes out 0, the value is a solution.

**Conjugate pairs:** when every coefficient is real, non-real solutions come in pairs  $a + bi$  and  $a - bi$  — so finding one hands you the other.

**Example:** Is  $x = 2 - i$  a solution of  $x^2 - 4x + 5 = 0$ ?

**Step 1.** Checking a proposed solution means substituting it and simplifying, not re-solving the equation.

**Step 2.**  $(2 - i)^2 = 3 - 4i$  and  $-4(2 - i) = -8 + 4i$ ; adding those with 5, the imaginary parts cancel and the real parts give  $3 - 8 + 5$ .

$$\Rightarrow 0, \text{ so yes — and } 2 + i \text{ is the other solution.}$$

**Try it:** Substitute  $x = 3 + i$  into  $x^2 - 6x + 10$ .

check: 0